JAL JEEVAN MISSION · PUBLIC INFRASTRUCTURE ANALYTICS

Har Ghar Jal
Jal Jeevan Mission

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FHTC Coverage & Performance Analysis

Interactive Data Analytics Dashboard using Microsoft Excel

NANDINI JAIN

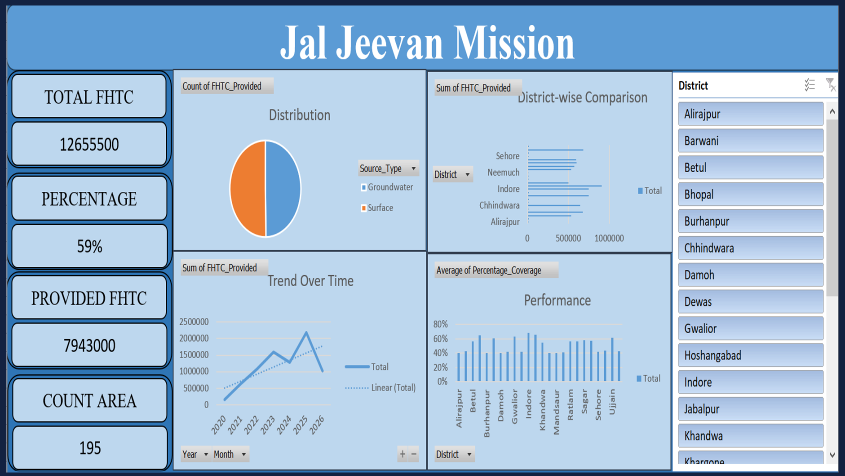
Data Analyst Portfolio

Excel

Data Analytics

Dashboard

Visualization



Excel Dashboard Preview

GETTING ORIENTED

Project Introduction

The Jal Jeevan Mission focuses on improving access to Functional Household Tap Connections (FHTCs) in rural India. This project analyzes FHTC implementation data to understand where progress stands and where attention is still needed.

Coverage

Scale of FHTC implementation achieved

FHTCs Provided

Volume of connections delivered

District Performance

How implementation compares across districts

Water-Source Distribution

Groundwater vs. surface-water reliance

Trends Over Time

How provision has moved year to year

This analysis translates raw implementation records into a single interactive view of program progress.

THE CHALLENGE

Problem Statement

The Jal Jeevan Mission aims to provide Functional Household Tap Connections to rural households. Monitoring implementation across districts requires an analytical view of coverage, performance, and water-source dependency.

KEY QUESTIONS

- **How much FHTC coverage has been achieved?**
- **Which districts are performing better or worse?**
- **How has FHTC provision changed over time?**
- **What is the distribution between groundwater and surface water?**
- **Which districts may require additional attention?**

From Data → Decisions

ANALYTICAL NEED

An interactive analytical dashboard can help convert implementation data into actionable insights for monitoring, comparison, and decision-making.

Raw Implementation Data

↓

Structured Analysis

↓

Interactive Dashboard

↓

Actionable Insight

ROADMAP

Project Objectives

01

Measure FHTC coverage

02

Analyze FHTCs provided

03

Compare district performance

04

Analyze groundwater vs. surface water

05

Track FHTC trends over time

06

Identify performance gaps

07

Build an interactive Excel dashboard

08

Generate data-driven recommendations

BEHIND THE SCENES

Data Cleaning & Preparation

Raw Data → Data Quality Check → Missing Value Check → Duplicate Validation → Standardization → Data Validation → Analysis-Ready Data

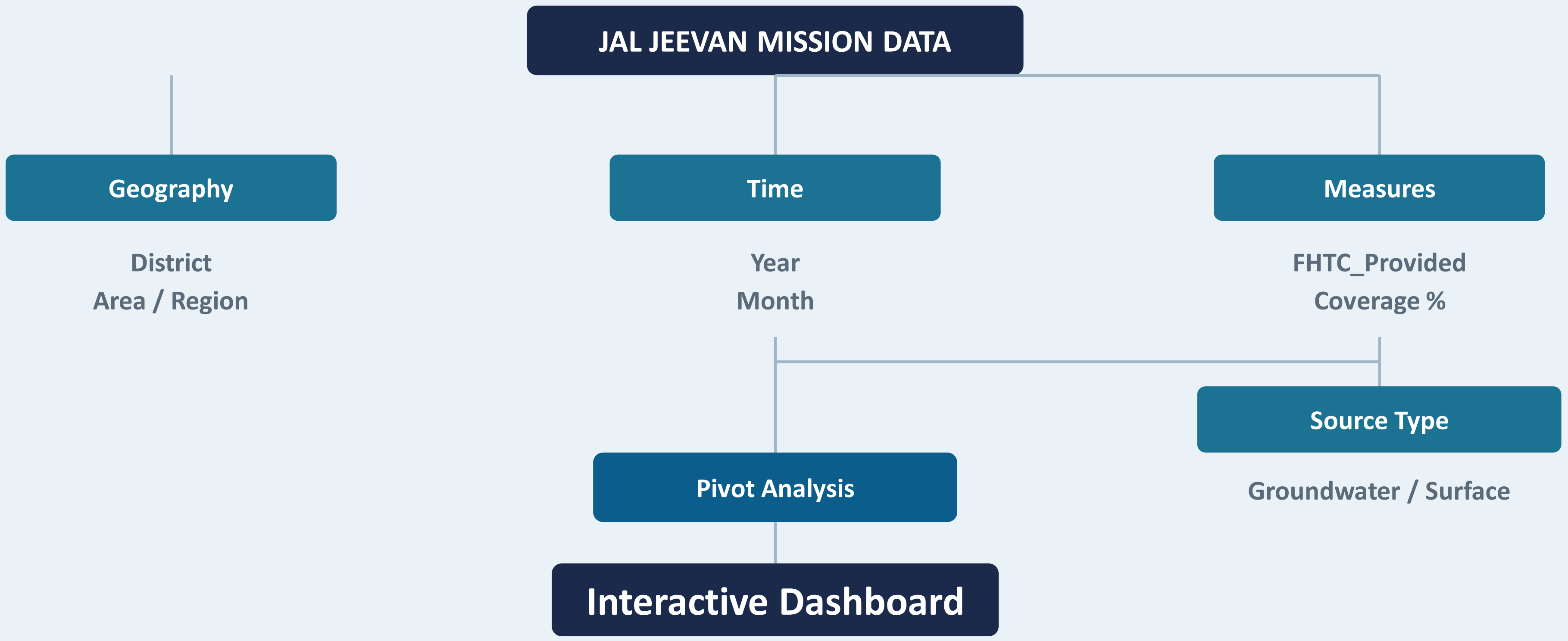
DATA PREPARATION ACTIVITIES

- Missing-value checks
- Data-type validation
- Numerical validation
- Preparation for pivot-table analysis
- Duplicate checks
- Standardization of categorical values
- Metric validation

These activities reflect the data-preparation methodology followed to make the dataset analysis-ready for pivot-based reporting.

BEHIND THE DASHBOARD

Data Modeling & Analytical Structure



This structure supports district analysis, time analysis, coverage analysis, source analysis, and dashboard reporting.

THE DATA

Dataset & Key Metrics

12.66M
Total FHTC

7.94M
FHTC Provided

59%
Coverage

195
Areas

ANALYTICAL DIMENSIONS

Geography
District / Area

Time
Year / Month

FHTC
FHTC Provided

Coverage
Percentage Coverage

Source
Groundwater / Surface

Figures sourced directly from the underlying dataset and Excel dashboard · No values altered or estimated

FRAMING THE ANALYSIS

Business Questions

Seven analytical questions guide the deep-dive analysis on the following slides.

01	What is the overall status of FHTC coverage?
02	How are FHTCs distributed by water source?
03	Which districts have the highest FHTC provision?
04	How has FHTC provision changed over time?
05	Which districts show stronger or weaker coverage performance?
06	Does higher FHTC provision necessarily correspond to higher coverage?
07	Which districts should receive greater monitoring attention?

BUSINESS QUESTION 1

"What is the overall status of Functional Household Tap Connection (FHTC) coverage?"

12.66M
Total FHTC

7.94M
FHTC Provided

195
Count Area

59.3%

AVERAGE FHTC COVERAGE

7,943,000 FHTCs provided out of 12,655,500 total FHTC requirement

KEY FINDING

59.3% Coverage

INSIGHT

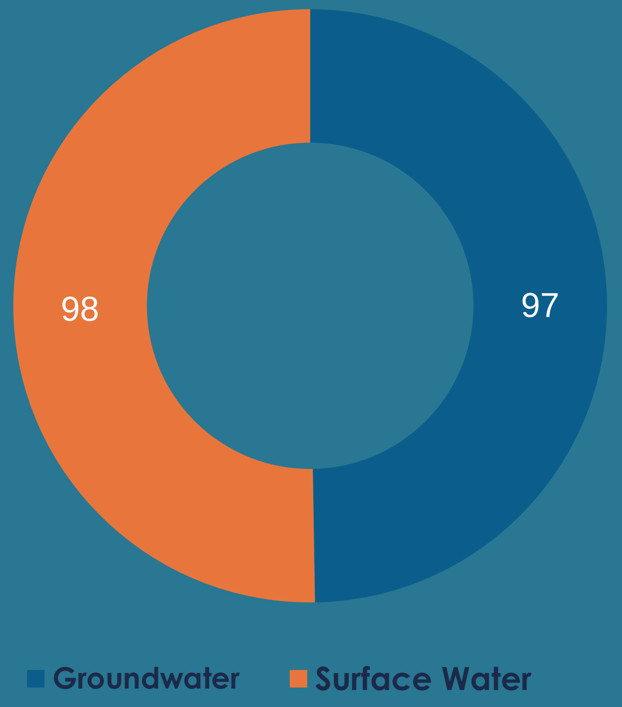
7.943 million FHTCs have been provided against 12.656 million total FHTCs, resulting in approximately 59.3% average coverage across the analyzed records.

BUSINESS IMPLICATION

While substantial progress has been made, approximately 40.7% of the total FHTC requirement remains uncovered, indicating scope for further expansion.

BUSINESS QUESTION 2

"How are FHTC records distributed between Groundwater and Surface Water sources?"



50.3% vs 49.7%

SURFACE WATER vs GROUNDWATER

Surface Water	98 records
Groundwater	97 records
Total	195 records

KEY FINDING

50.3% Surface vs 49.7% Groundwater

INSIGHT

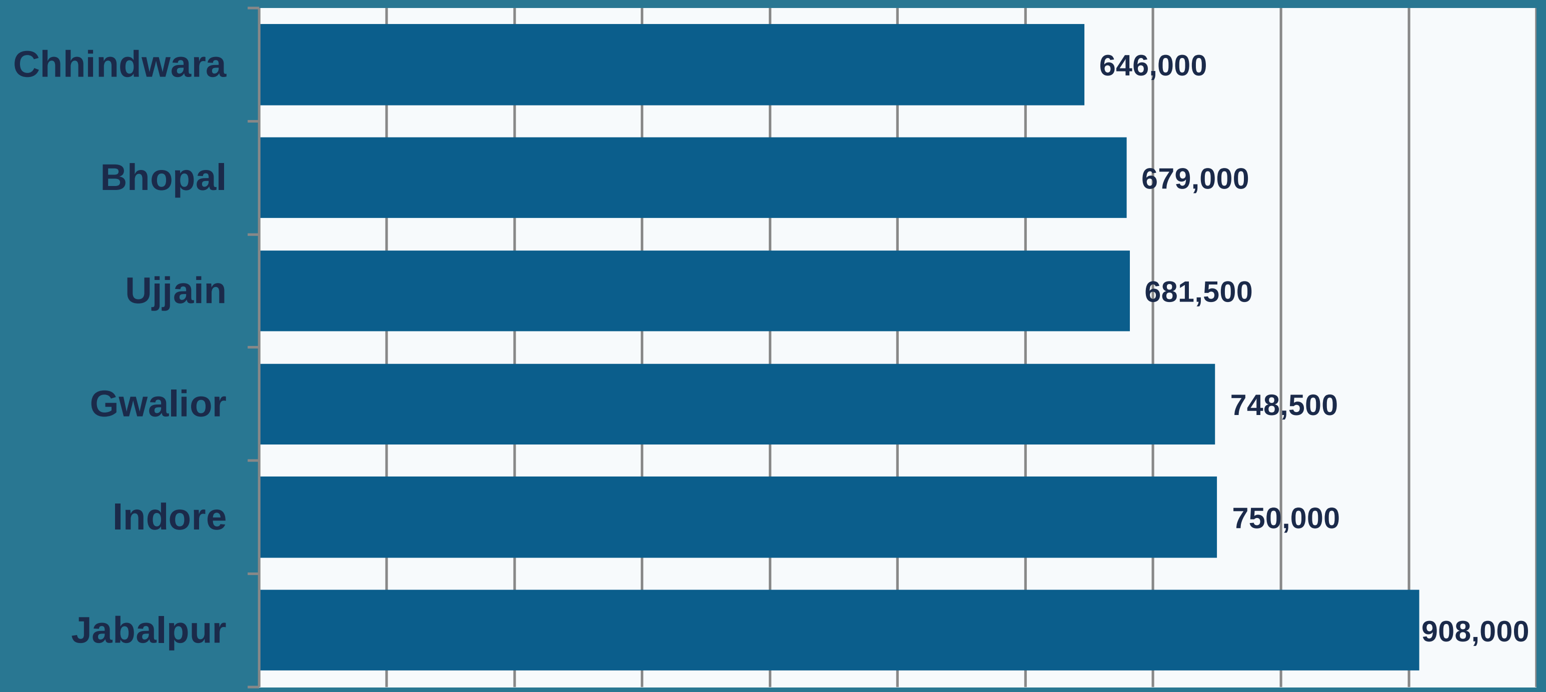
The source distribution is almost evenly split, with 98 Surface Water records and 97 Groundwater records.

BUSINESS IMPLICATION

The near-balanced source mix indicates that implementation is utilizing both groundwater and surface-water sources rather than relying predominantly on one source category.

BUSINESS QUESTION 3

"Which districts have the highest FHTC provision, and how significant is the variation across districts?"



Jabalpur

908K FHTCs Provided

Highest FHTC provision across all districts

KEY FINDING

Jabalpur — 908K FHTCs Provided

INSIGHT

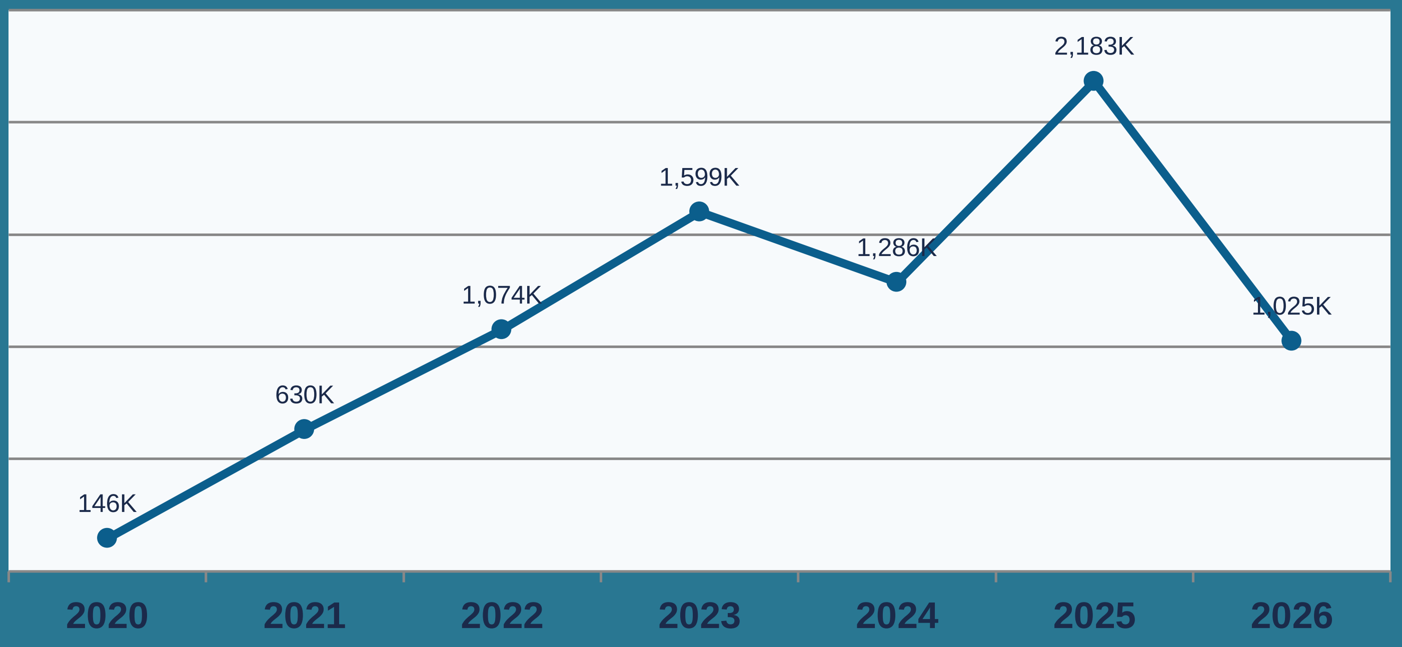
Jabalpur has the highest FHTC provision at 908,000, followed by Indore at 750,000 and Gwalior at 748,500.

BUSINESS IMPLICATION

District-level variation highlights the need for differentiated monitoring and resource planning rather than applying a uniform implementation strategy across all districts.

BUSINESS QUESTION 4

"How has FHTC provision changed over the years?"



2.183M

PEAK YEAR: 2025

Highest displayed FHTC provision in a single year

KEY FINDING

2025 — Peak at 2.183M

INSIGHT

FHTC provision increased substantially from 146K in 2020 to 1.599M in 2023, declined in 2024, reached the highest displayed value of 2.183M in 2025, and then decreased to 1.025M in 2026.

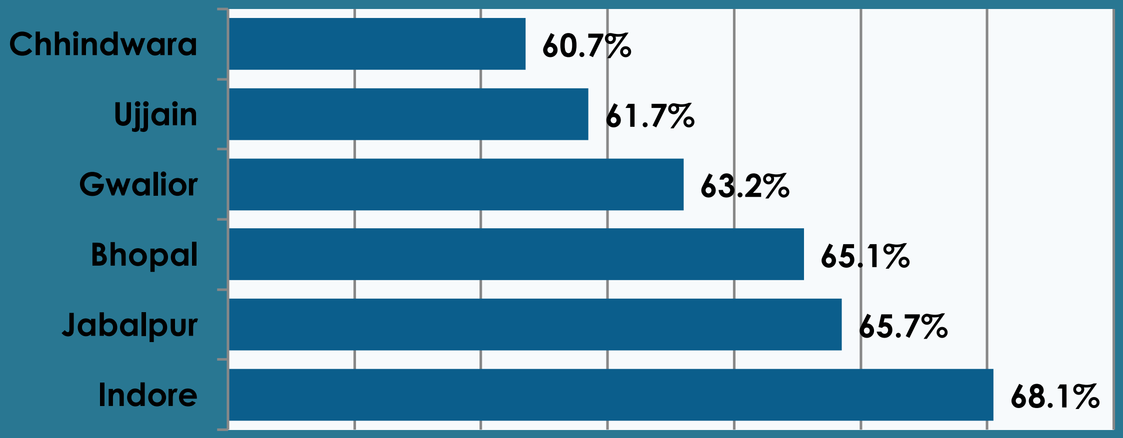
BUSINESS IMPLICATION

The fluctuations suggest that implementation progress should be monitored periodically to identify changes in the pace of FHTC provision.

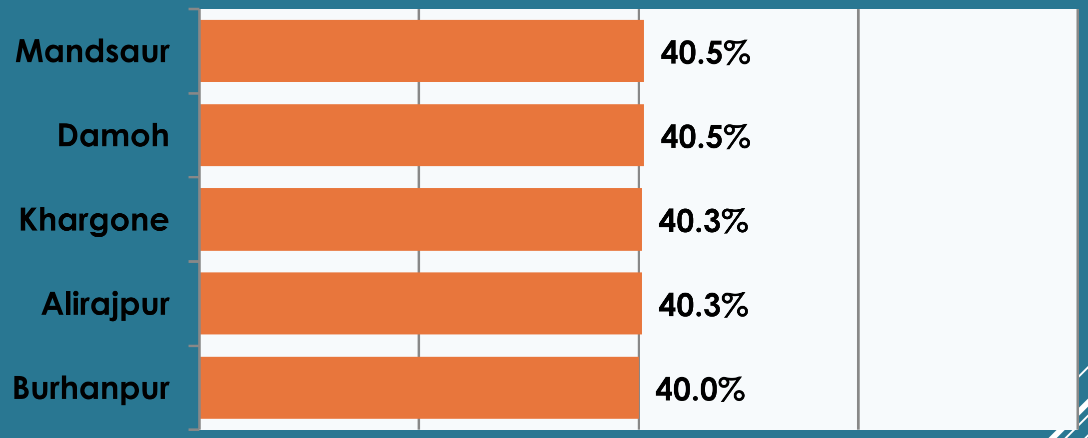
BUSINESS QUESTION 5

"Which districts demonstrate stronger and weaker average coverage performance?"

HIGHER-PERFORMING DISTRICTS



LOWER-PERFORMING DISTRICTS



Indore — Highest Average Coverage: 68.1%

KEY FINDING

Indore — 68.1% Coverage

INSIGHT

Indore has the highest average coverage at approximately 68.1%, while several districts remain around the 40–43% range.

BUSINESS IMPLICATION

Lower-coverage districts can be prioritized for closer monitoring, while high-performing districts can provide useful benchmarks for implementation practices.

BUSINESS QUESTION 6

"Does higher FHTC provision necessarily correspond to higher coverage performance across districts?"

VOLUME ≠ COVERAGE

Jabalpur

Higher Volume

FHTC Provided: **908,000**

Avg. Coverage: **65.7%**

Indore

Higher Coverage

FHTC Provided: **750,000**

Avg. Coverage: **68.1%**

INSIGHT

Absolute FHTC provision and percentage coverage do not necessarily tell the same story. Jabalpur has a higher FHTC provision than Indore, while Indore has a higher average coverage.

BUSINESS IMPLICATION

Performance monitoring should consider both the scale of implementation and coverage efficiency rather than relying on FHTC volume alone.

BUSINESS QUESTION 7

"Which districts should receive greater monitoring attention based on coverage performance?"

Priority Group

Lower Coverage Districts

Substantially below the overall average coverage of approximately 59.3%

Burhanpur	40.0%
Khargone	40.3%
Mandsaur	40.5%
Hoshangabad	42.1%
Alirajpur	40.3%
Damoh	40.5%
Neemuch	41.3%
Sehore	42.1%

KEY FINDING

Priority Group Identified

INSIGHT

Several districts have average coverage around 40–43%, substantially below the overall average coverage of approximately 59.3%.

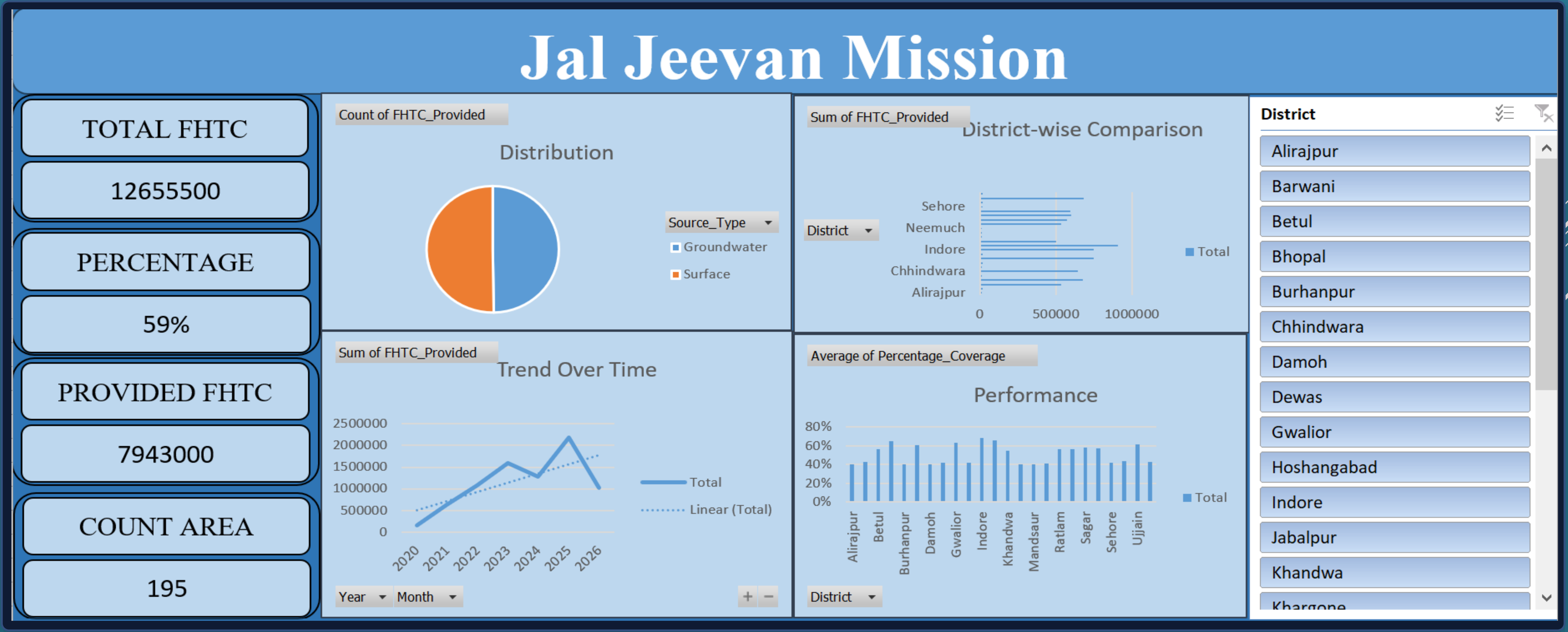
BUSINESS IMPLICATION

These districts can be considered priority areas for deeper investigation, targeted implementation monitoring, and resource planning.

THE DELIVERABLE

Interactive Excel Dashboard

One dashboard. Multiple analytical perspectives.



KPI Cards | Source Distribution | District Comparison | Trend Analysis | Coverage Performance | District Slicer

KEY RESULTS

Executive Summary

01

59.3%

Average coverage

02

7.943M

FHTCs provided

03

908K

Jabalpur — highest displayed FHTC provision

04

68.1%

Indore — highest average coverage

05

2.183M

2025 — highest displayed FHTC provision

06

40–43%

Several districts remain in this coverage range

07

≈50/50

Groundwater and surface water nearly evenly split

INSIGHT → ACTION

Data-Driven Recommendations

01

Prioritize Lower-Coverage Districts

Focus monitoring and intervention on districts with comparatively lower coverage.

02

Monitor Regional Performance Gaps

Investigate differences in implementation performance across districts.

03

Learn from High-Performing Districts

Study high-performing districts and identify practices that may be adapted elsewhere.

04

Monitor Water-Source Mix

Track groundwater and surface-water utilization as part of implementation monitoring.

05

Strengthen Trend Monitoring

Review monthly / yearly changes to identify fluctuations in implementation.

06

Use Dashboard for Periodic Reviews

Use the dashboard for recurring district-level performance monitoring and decision-making.

WRAPPING UP

Conclusion

The Jal Jeevan Mission dashboard transforms FHTC implementation data into a consolidated view of coverage, district performance, source distribution, and trends over time.

- Overall progress monitoring
- District comparison
- Coverage performance analysis
- Trend analysis
- Source distribution monitoring
- Data-driven prioritization

From Raw Data →
Analysis →
Dashboard →
Insights → Action

CAPABILITIES

Skills Demonstrated

Data Analytics

- Data Cleaning
- Exploratory Data Analysis
- KPI Analysis
- Trend Analysis
- Comparative Analysis

Excel

- Pivot Tables
- Pivot Charts
- Slicers
- Calculations
- Dashboard Development

Business Intelligence

- Data Visualization
- Insight Generation
- Data Storytelling
- Performance Monitoring
- Recommendations

Domain

- Government Analytics
- Public Infrastructure
- Water & Rural Development

Turning Data into Decisions.